Graphical SLAM

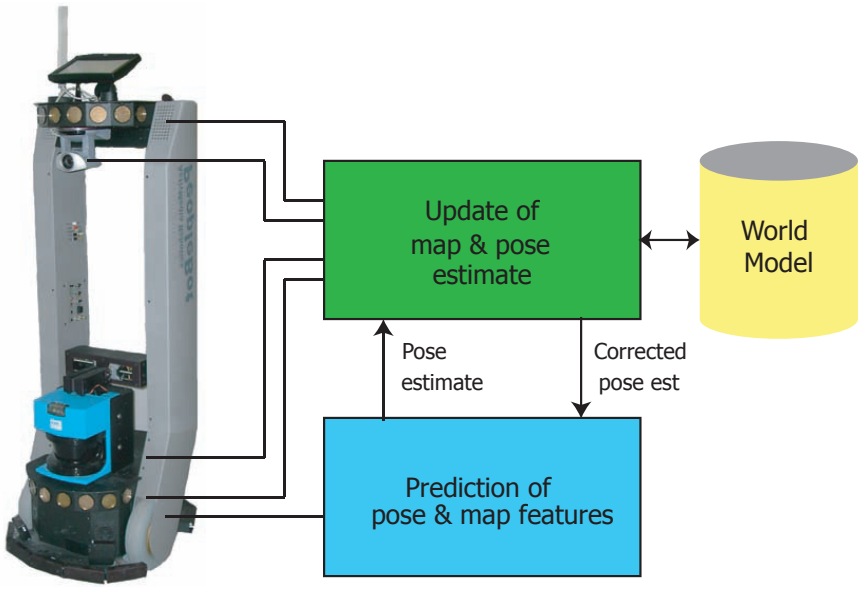
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Outline

1. Introduction
2. The Problem
3. Graph Model
4. Status?
5. Graph reduction
6. Loop closing
7. Example
8. Summary

Outline of problem



Ensure the robot does not get lost!

The basics for SLAM

- State of robot is modelled as the pose

$$\vec{x}_R = \begin{bmatrix} x & y & \theta \end{bmatrix}^T$$

- Map features can be represented as points or lines, i.e.:

$$\vec{x}_i = \begin{bmatrix} x_i & y_i \end{bmatrix}^T$$

Estimation as a Kalman Problem

- Prediction by odometric modelling
- Updating as a Kalman process, with the state

$$\vec{x}_{state} = \begin{bmatrix} \vec{x}_R \\ \hline \vec{x}_1 \\ \vdots \\ \vec{x}_n \end{bmatrix}$$

Why is SLAM difficult?

- The number of map hypotheses is very large
- Often the signal to noise ratio for features is ≈ 1
- Robust discriminative features are not common
- The "process" is often approximated

Problems?

1. Flexible inclusion/exclusion of measurements?
2. Handling of linearization?
3. Dealing with topological constraints?
   - Loop closing etc.

Data handling

- Easy inclusion and/or exclusion of data at any time in the process.
- How to avoid too early a commitment to a particular map hypothesis.
- Design of a representation that allow any-time inclusion/exclusion of data?

Linearizations?

- Linearization might cause divergence in the data.
- Reported by several, e.g. Julier and Uhlmann [2001], Castellanos et al. [2004]
- Consistent handling of non-linearities
  - Start by exact handling of non-linearities
  - As data matures a linearization is permitted
  - Identification of major non-linearities to include them

Topological constraints?

- Consistent inclusion of topological constraints
- Two step strategy:
  1. Close approximation of system in a trivial way
  2. Fine tune full model by adding "smaller" corrections

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Problem statement

- $\{x_i\}$ the robot path (set of poses), $(i \in \{1 \ldots N_p\})$
- $\{z_j\}$ feature coordinates $(j \in \{1..N_m\})$
- $\{d_i\}$ dead reckoning measurements, between feature measurements
- $\{f_k\}$ feature measurements, $(k \in \{1..N_f\})$
- Λ the $f \leftrightarrow z$ association

$$P(x, z, d, f, \Lambda) = P(d, f|x, z, \Lambda)P(x, z, \Lambda)$$

Probabilistic model

$$
\begin{aligned}
P(x, z, d, f, \Lambda) &\propto P(d, f|x, z, \Lambda)P(x, z, \Lambda) \\
P(d, f|x, z, \Lambda) &\propto P(d|x)P(f|x, z, \Lambda) \\
P(x, z, \Lambda) &\propto P(\lambda) = P(N_f) \propto e^{-\lambda N_f} \\
P(x, z, d, f, \Lambda) &\propto P(d|x)P(f|x, z, \Lambda)e^{-\lambda N_f}
\end{aligned}
$$

An energy model

- Definition of energy/entropy of the model:

$$E(x, z, d, f, \Lambda) = -\log(P(d|x)) - \log(P(f|x, z, \Lambda)) + \lambda N_f$$

- Or $E(x, z, d, f, \Lambda) = E_d + E_f + E_\Lambda$
- Or: ...

An energy model

$$E(x, z, d, f, \Lambda) = E_d(x) + E_f(x, z) + E_\Lambda(n_j) \quad (1)$$

$$E_d = -\sum_{i=1}^{N_p} log(P(d_i|x_{i-1}, x_i)) = \frac{1}{2}\sum_{i=1}^{N_p} \xi_i^T k_i \xi_i \quad (2)$$

$$E_f = -\log(P(f|x, z, \Lambda)) = \frac{1}{2}\sum_{k=1}^{N_m} \eta_k^T k_k \eta_k \quad (3)$$

$$E_\Lambda = -\sum_{j=1}^{N_f} \lambda(n_j - 1) \quad (4)$$

$$\xi_i = T(x_i|x_{i-1}) - d_i \qquad \eta_k = h(T(z_j|x_i)) - f_k$$

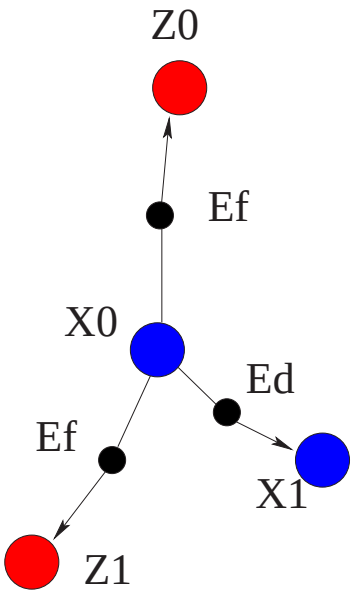
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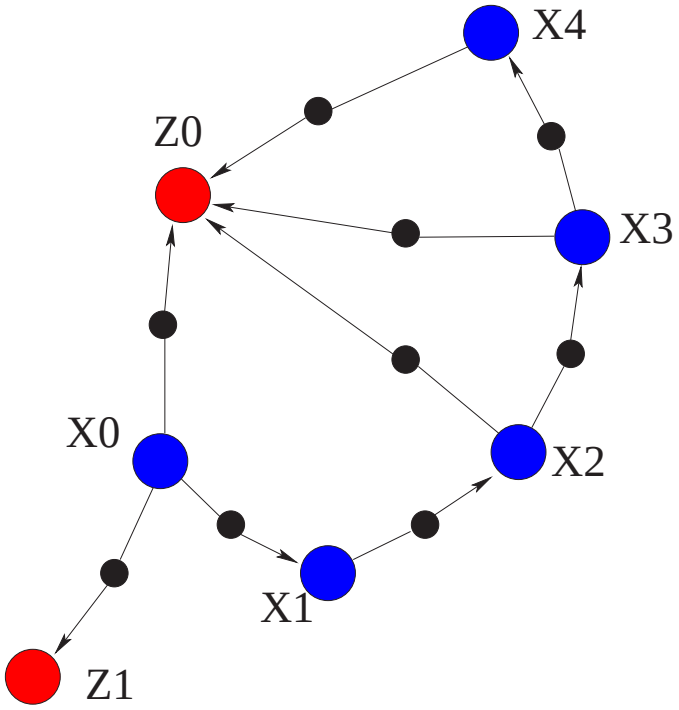
Organizing a model

- Graph representation
- Two types of nodes:
  1. State notes
     - Poses (x_i)
     - Features (z_j)
  2. Energy Nodes (Computation of Eqn (1))
     - Connected to the state nodes needed for computation
     - Movement (d_i, k_i)

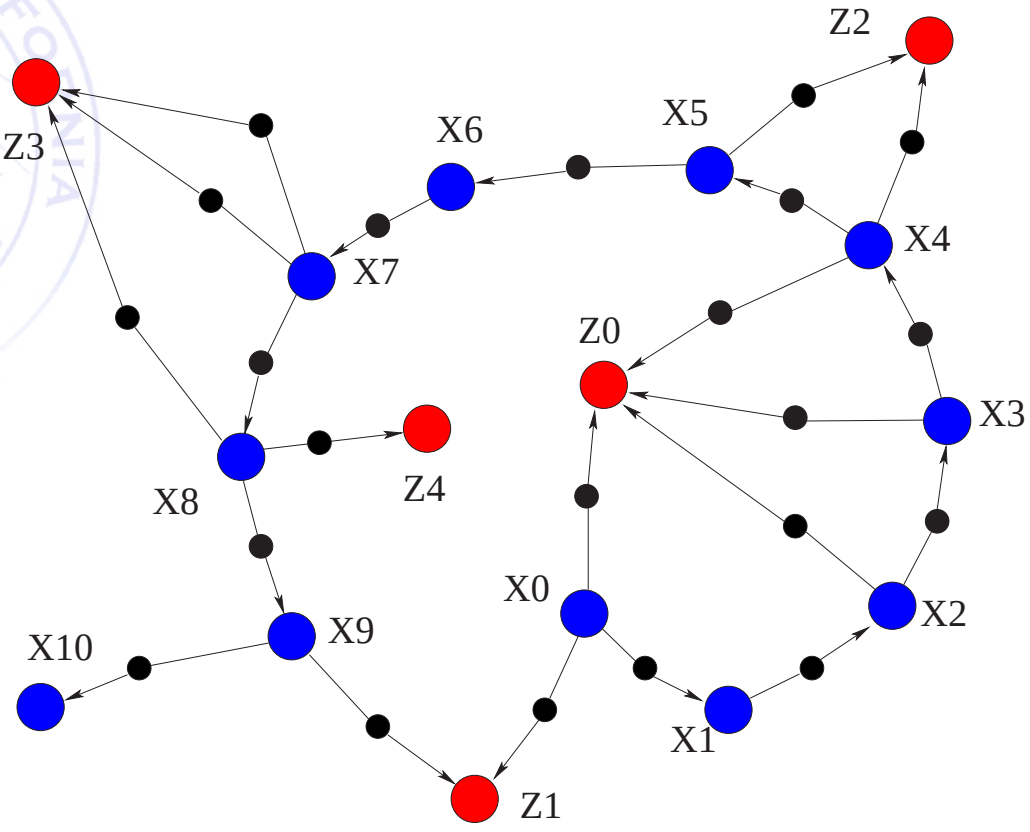
Starting a model



Entering more data



A graphical model example



Map updating

- Optimal solution to Eq. (1): $(\mathrm{argmin}E)$ is not realistic.
- Relaxation techniques allow iterative updating
- In a time step:
  1. Add a new state node (pose)
  2. Any new features/measurements?
  3. Update the rest of the map – minimize energy

Map Update–Energy

- Eqn (1) Taylor expanded for a node A:

$$
\begin{aligned}
E_A &= \sum_{i \in edge(A)} [E_i(\bar{x}_A, \bar{x}_i) + \nabla E_i(\bar{x}_A, \bar{x}_i) \begin{pmatrix} \Delta x_A \\ \Delta x_i \end{pmatrix} \\
&+ \frac{1}{2} \begin{pmatrix} \Delta x_A & \Delta x_i \end{pmatrix} \nabla^T \nabla E_i(\bar{x}_A, \bar{x}_i) \begin{pmatrix} \Delta x_A \\ \Delta x_i \end{pmatrix}]
\end{aligned}
$$

Notation:

$$
\begin{aligned}
\mathcal{G} &= \nabla E_A = \begin{pmatrix} \mathcal{G}_A \\ \mathcal{G}_i \end{pmatrix} \\
\mathcal{H} &= \nabla^T \nabla E_A = \begin{pmatrix} \mathcal{H}_{AA} & \mathcal{H}_{Ai} \\ \mathcal{H}_{Ai} & \mathcal{H}_i i \end{pmatrix}
\end{aligned}
$$

Map Update–Energy (II)

- Optimizing a node A:

$$(x_A - \bar{x}_A) = -\mathcal{H}_{AA}^{-1}\mathcal{G}_A \tag{5}$$

- With an energy change of

$$\Delta E_A = \frac{1}{2}\mathcal{G}_A^T\mathcal{H}_{AA}^{-1}\mathcal{G}_A \tag{6}$$

Map Update–Energy (III)

- ΔE_A decides on update strategy:
  - Steepest decent – close to saddle point
  - Use eqn. (5) direct
  - Chained update
    - Locate a node that is "good"
    - Update from that node

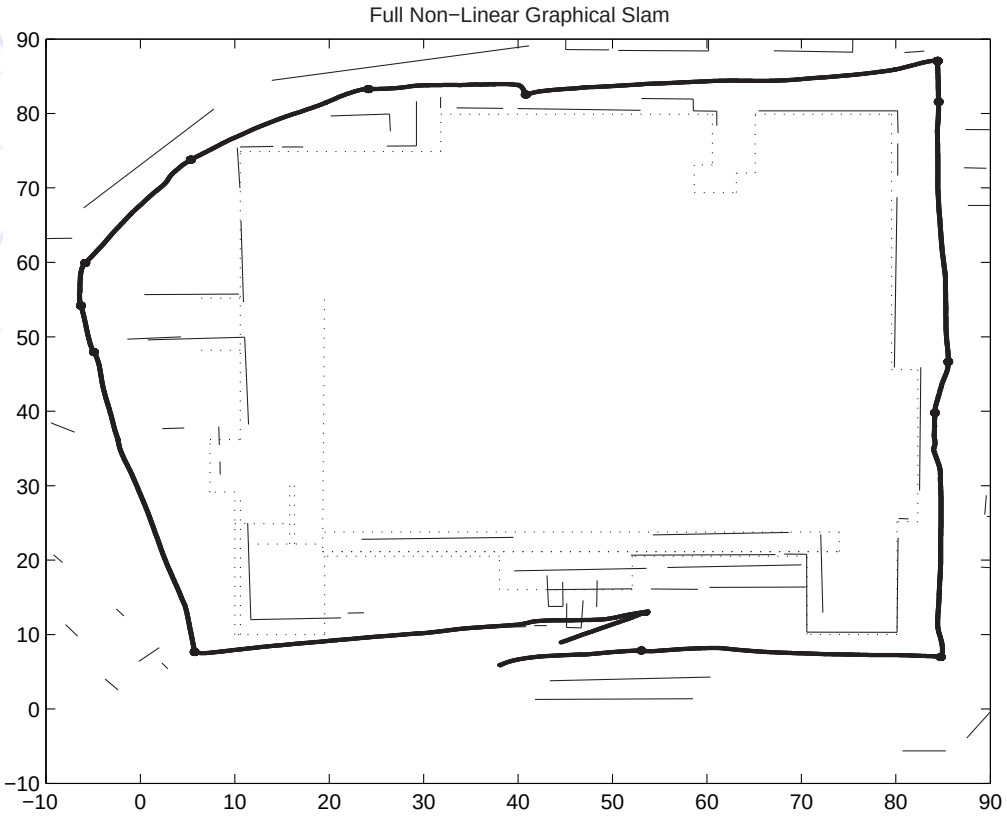
Feature Matching

- λ decides on the value of new measurements
  1. Add a new feature
  2. Compute energy change
  3. If change too large, remove association
- Similar to EM [Liu et al., 2001]
- Matching/graph updating anytime

Example

- ATRV Vehicle with Trimble DGPS 212
- Sick LMS 291 Laser scanner
- CrossBow DFOG INS package
- State $(\theta_n, x_n, y_n)^T$
- Lines w. end-points (x, y)
- See Folkesson and Christensen [2003] for details
- Operating around a house.
- 7500 Pose Nodes, 11975 line measurements
- Update time 30 ms (550 MHz Pentium)

SLAM example



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Status?

- Flexible association of data, anytime
- A framework to avoid linearization effects
- Loop closing:
  1. Recognition of loop closure (place recognition)
  2. Updating map to include the topological constraint
- Doing 2) often has limited effect
- A proposal . . .

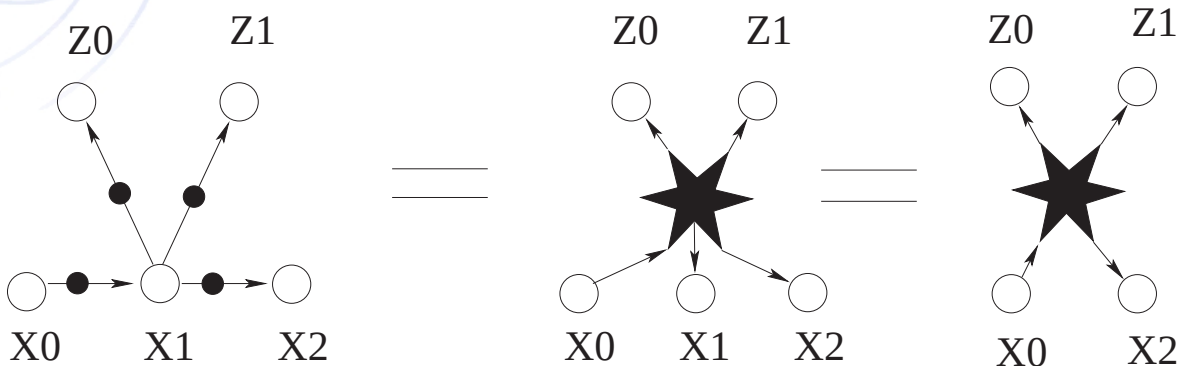
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Reducing the graph

- How can state nodes be removed from the graph?
- In a linear system one could generate a closed form solution
- In a non-linear case an approximation can be used.
- Consider elimination of a node A given neighboring nodes B.
- Consider the energy from state A expanded.

The basic idea



The Energy from A

$$
\begin{aligned}
E_A &= \sum_{j \in edge(A)} [E_i(\bar{x}_A, \bar{x}_j) + \mathcal{G}_j(x_j - \bar{x}_j) \\
&+ \frac{1}{2} \begin{pmatrix} x_A - \bar{x}_A & x_j - \bar{x}_j \end{pmatrix} \mathcal{H}_j(\bar{x}_A, \bar{x}_j) \begin{pmatrix} x_A - \bar{x}_A \\ x_j - \bar{x}_j \end{pmatrix}] \qquad (7)
\end{aligned}
$$

Transforming to B:

$$(x_A - \bar{x}_A) = -\mathcal{H}_{AA}^{-1} \mathcal{H}_{AB} (x_B - \bar{x}_B)$$

Elimination of A

- Substitution into eqn (7) eliminates A

$$
\begin{aligned}
E^* &= \sum_{j \in edge(A)} [E_j(\bar{x}_A, \bar{x}_j) + \mathcal{G}_j(x_j - \bar{x}_j) \\
&+ \frac{1}{2} (x_j - \bar{x}_j)^T \mathcal{H}_{jj} (x_j - \bar{x}_j) \\
&- \sum_{i \in edge(A)} (x_i - \bar{x}_i)^T \mathcal{H}_{iA} \mathcal{H}_{AA}^{-1} \mathcal{H}_{Aj} (x_j - \bar{x}_j)]
\end{aligned}
$$

- New term connects i and j (new connection)

Star nodes

- The fused nodes are termed *Star* nodes.
- In principle the entire graph could be fused into a single node, as done in EKF.
- In practice star nodes are used to fuse local "maps"
- Typically upto 128 state nodes are considered for "fusion".
- The star nodes are considered retroactively after say 50 poses.

Making star nodes invariant

- Star nodes updates energy using the Hessian $\mathcal{H}$
- If measurements have symmetries (such as lines) $\mathcal{H}$ will have zero eigenvalues.
- Project state nodes to "natural coordinates" (q) in a lower dimensional space without symmetries.

$$q_i = PT(x_i|x_0)$$

Mapping to natural coordinates

- In general:

$$\mathcal{H}_{xx} = J^T \mathcal{H}_{qq} J + \frac{\partial^2 q}{\partial x \partial x} \mathcal{G}_q \approx J^T \mathcal{H}_{qq} J$$

- Thus:

$$\mathcal{H}_{qq} = \tilde{J} \mathcal{H}_{xx} \tilde{J}^T$$

- Using SVD it is possible to find eigenvalues b_j and eigenvectors V_j of the Hessian

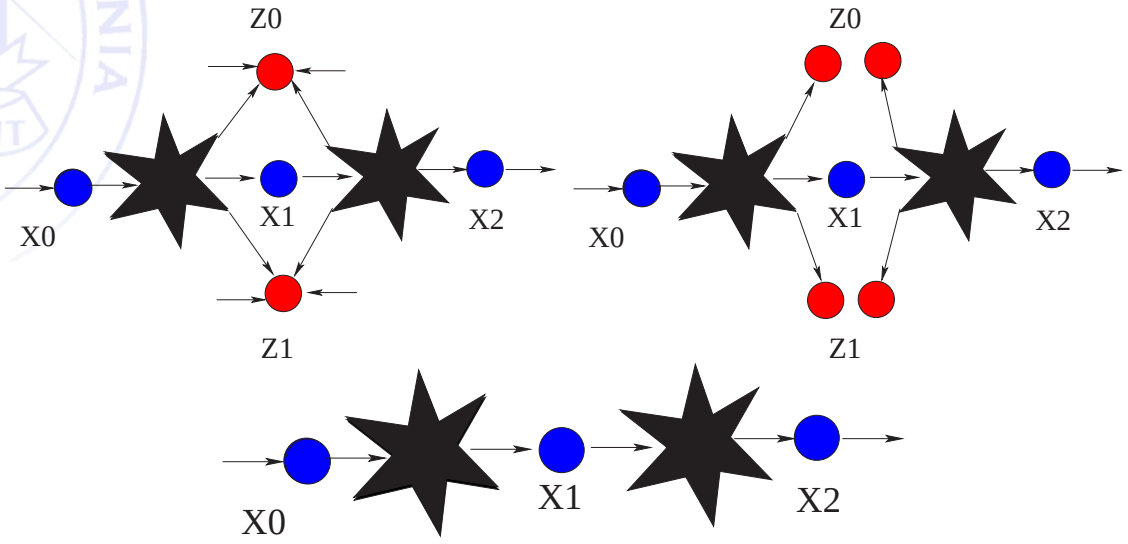
Energy & Properties

- Using the eigenvectors the energy at $\bar{q}$ is now:

$$E^* = E^*(\bar{q}) + \frac{1}{2} \sum_j b_j (V_j \Delta q)^2$$

- Recentered nodes are linear in energy
- A state node connected to only one star node can be eliminated ($\Delta q = 0$)
- Star nodes like local maps in Atlas [Bosse et al., 2003]

Optimizing global calculations



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Closing loops

- With $\Delta q = 0$ the state nodes between star nodes is ignored.
- Define a cost function using Lagrange multipliers

$$C(\Lambda, \Delta q) = \Lambda(\sum_i \Delta x - d_c) + \frac{1}{2} \sum_i \sum_j b_{ji} (V_{ji} \Delta q)^2$$

- Where i is the star node index, Δx_i is the pose difference between the poses. d_c is the pose constraint.
- I.e.

$$\Delta x_i = \begin{pmatrix} \mathcal{R}_i & 0 \\ 0 & 1 \end{pmatrix} (\Delta q_i + \bar{q}_i)$$

Solving for closed loops

- Linearisation gives:

$$\Delta q_i = -\sum_j \frac{V_{ji}^T V_{ji}}{b_j} \begin{pmatrix} \mathcal{R}_i & 0 \\ 0 & 1 \end{pmatrix} \Lambda^T$$

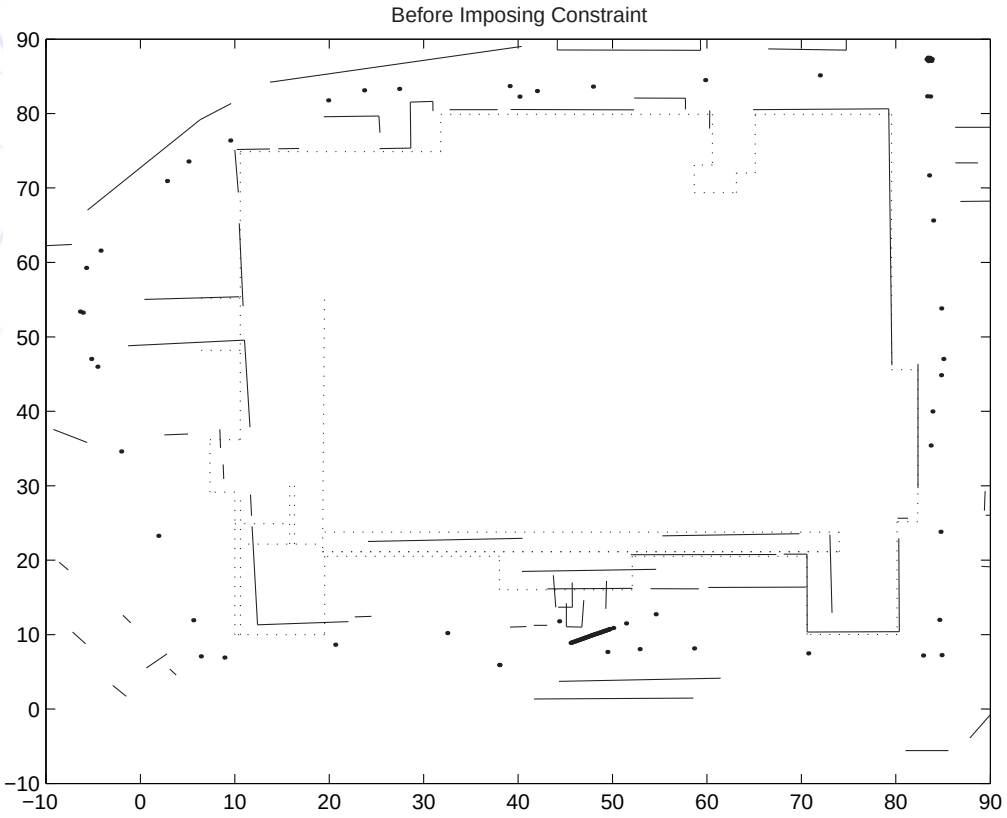
- Where:

$$\begin{aligned} \Lambda^T &= S^{-1} \left\{ \left[\sum_i \begin{pmatrix} \mathcal{R}_i & 0 \\ 0 & 1 \end{pmatrix} \bar{q}_i \right] - d_c \right\} \\ S &= \sum_i \left\{ \begin{pmatrix} \mathcal{R}_i & 0 \\ 0 & 1 \end{pmatrix} \left[\sum_j \frac{V_{ji}^T V_{ji}}{b_j} \right] \begin{pmatrix} \mathcal{R}_i & 0 \\ 0 & 1 \end{pmatrix} \right\} \end{aligned}$$

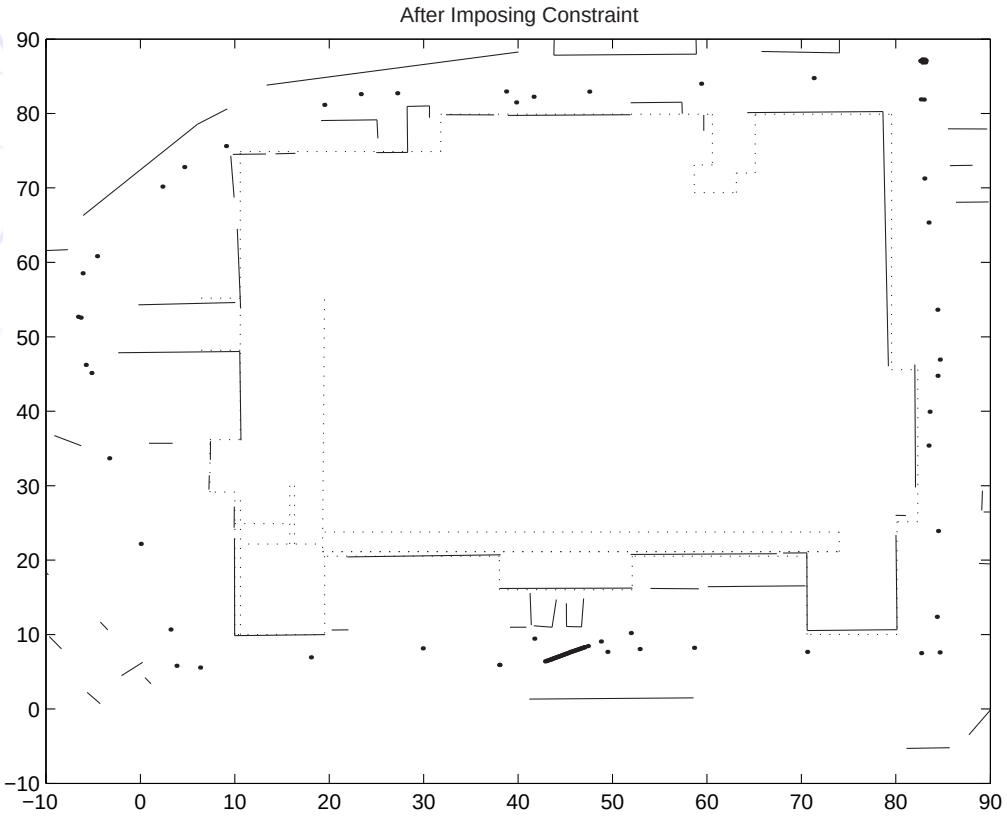
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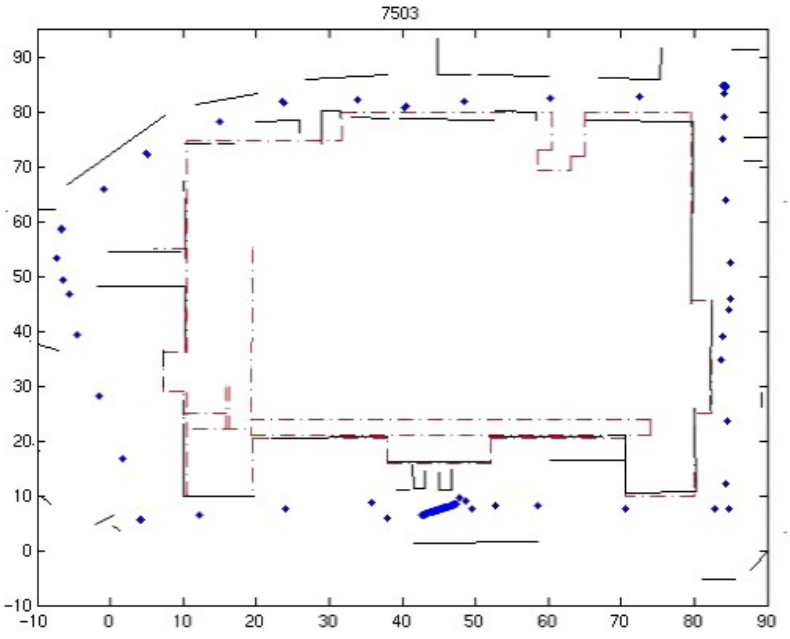
Map example - no closing



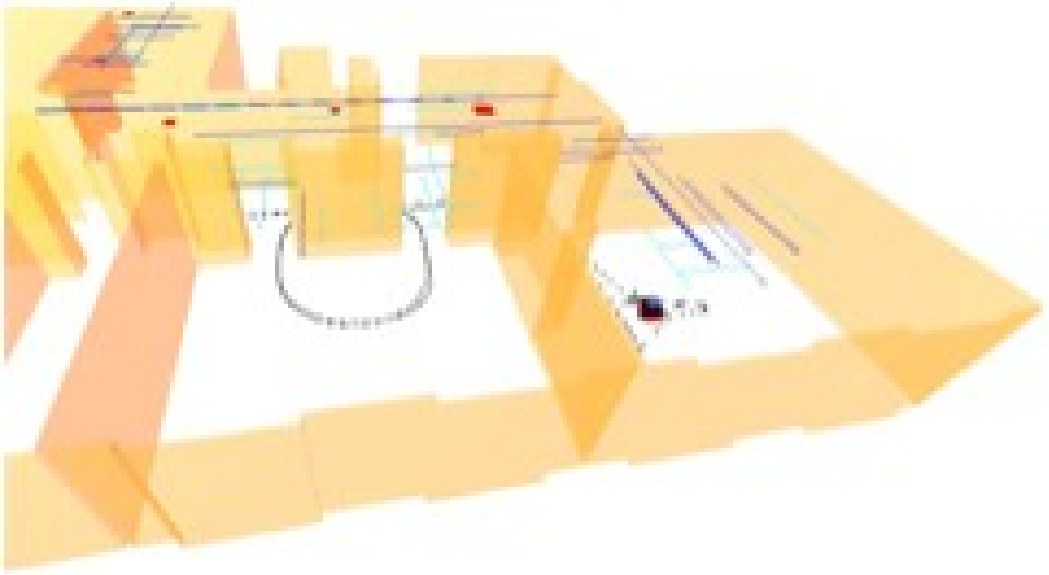
Map example - closing



Example Movie



Example Movie 2



Closing loops

- Loop closing has a complexity of $O(N+1)$ where N is the number of star nodes.
  - Once loop closing is achieved
  - Turn back on inter-star relation to fine tune
- Achieved in 1-2 seconds for large environments
- Constraints similar to "strong links" in Lu and Milios [1997].

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Summary

- A new efficient representation for SLAM
  - Easy integration of data anytime
  - "Chunk-ing" data into local maps (star nodes)
  - Addition of constraint for loop closing
- Computationally efficient (10-12 ms / pose)
- Losing closing without linearization problems
- Further details in Folkesson and Christensen [2004b,a], Thrun et al. [2001, 2006]

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